

Foundation (Jalapeno)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** If two lines have slopes $m_1 = 3$ and $m_2 = -\frac{1}{3}$, what can be concluded about the lines?
(A) They are parallel
(B) They are perpendicular
(C) They are skewed
(D) They are vertical
(E) They are horizontal
- Q3.** A basketball player shoots a ball from the free throw line. The line from the free throw line to the basket is modeled by $y = 4x - 2$. What is the equation of a line perpendicular to this line that passes through the point (2, 2)?
(A) $y = -\frac{1}{4}x + \frac{5}{2}$
(B) $y = -\frac{1}{4}x + \frac{3}{2}$
(C) $y = 4x + 2$
(D) $y = -4x + 2$
(E) $y = \frac{1}{4}x - \frac{3}{2}$
- Q4.** A tennis player serves the ball from the baseline. The line from the baseline to the opponent's court is modeled by $y = -2x + 5$. What is the slope of a line perpendicular to this line?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q5.** A cyclist is riding on a road with a slope of $m = \frac{3}{4}$. What is the slope of a road perpendicular to this road?
(A) $-\frac{4}{3}$
(B) $\frac{4}{3}$
(C) 3
(D) -3
(E) 0
- Q6.** A runner is running on a track with a slope of $m = -\frac{2}{5}$. What is the equation of a line perpendicular to this line that passes through the point (1, 3)?
(A) $y = \frac{5}{2}x - \frac{1}{2}$
(B) $y = \frac{5}{2}x + \frac{1}{2}$
(C) $y = -\frac{5}{2}x + 3$
(D) $y = -\frac{5}{2}x + 2$
(E) $y = \frac{2}{5}x + 2$
- Q7.** A football player kicks the ball from the 50-yard line. The line from the 50-yard line to the opponent's end zone is modeled by $y = -x + 10$. What is the equation of a line perpendicular to this line that passes through the point (4, 2)?
(A) $y = x - 2$
(B) $y = -x - 2$
(C) $y = x + 2$
(D) $y = -x + 2$
(E) $y = x + 6$
- Q8.** A golfer hits the ball from the tee box. The line from the tee box to the green is modeled by $y = 2x - 1$. What is the slope of a line perpendicular to this line?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0

Open Ended Questions (FRQ)

- Q9.** A swimmer is swimming in a pool with a length of 25 yards. The line from one end of the pool to the other is modeled by $y = 3x - 2$. Find the equation of a line perpendicular to this line that passes through the point (5, 1), and then find the distance from the point (5, 1) to the line $y = 3x - 2$. Show your work and provide the final answer.
- Q10.** A volleyball player serves the ball from the service line. The line from the service line to the opponent's court is modeled by $y = -\frac{3}{2}x + 4$. Find the equation of a line perpendicular to this line that passes through the point (2, 3), and then find the y -intercept of the line $y = -\frac{3}{2}x + 4$. Show your work and provide the final answer.

Chef's Tips

- Emphasize the importance of identifying the negative reciprocal slope of a line to find the slope of a perpendicular line.
- Encourage students to use graph paper to visualize the lines and their relationships.
- Remind students to label their work and provide clear explanations for their answers.